

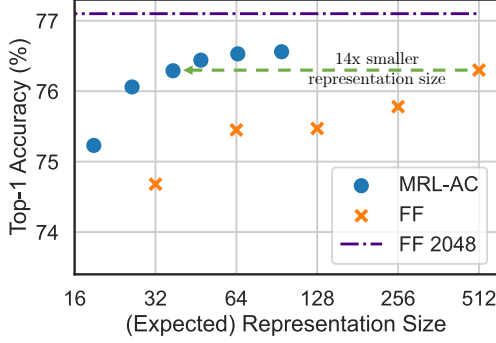


Figure 6: Adaptive classification on MRL ResNet50 using cascades results in $14\times$ smaller representation size for the same level of accuracy on ImageNet-1K (~ 37 vs 512 dims for 76.3%).

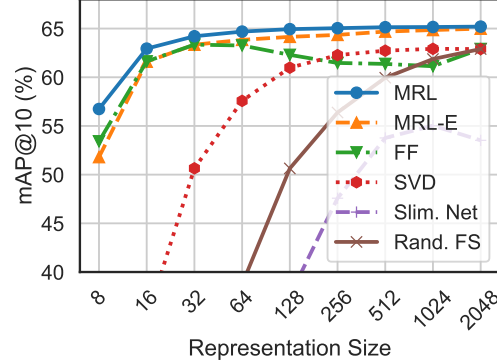


Figure 7: mAP@10 for Image Retrieval on ImageNet-1K with ResNet50. MRL consistently produces better retrieval performance over the baselines across all the representation sizes.

an extensive set of metrics spanning $\text{mAP}@k$ and $\text{P}@k$ for $k = \{10, 25, 50, 100\}$ and real-world wall-clock times for exact search and HNSW. See Appendices E and F for more details.

Figure 7 compares the $\text{mAP}@10$ performance of ResNet50 representations on ImageNet-1K across dimensionalities for MRL, MRL-E, FF, slimmable networks along with post-hoc compression of vectors using SVD and random feature selection. Matryoshka Representations are often the most accurate while being up to 3% better than the FF baselines. Similar to classification, post-hoc compression and slimmable network baselines suffer from significant drop-off in retrieval $\text{mAP}@10$ with ≤ 256 dimensions. Appendix E discusses the $\text{mAP}@10$ of the same models on ImageNet-4K.

MRL models are capable of performing accurate retrieval at various granularities without the additional expense of multiple model forward passes for the web-scale databases. FF models also generate independent databases which become prohibitively expensive to store and switch in between. Matryoshka Representations enable adaptive retrieval (AR) which alleviates the need to use full-capacity representations, $d = 2048$, for all data and downstream tasks. Lastly, all the vector compression techniques [60, 45] used as part of the ANNS pipelines are complimentary to Matryoshka Representations and can further improve the efficiency-vs-accuracy trade-off.

4.3.1 Adaptive Retrieval

We benchmark MRL in the adaptive retrieval setting (AR) [50]. For a given query image, we obtained a shortlist, $K = 200$, of images from the database using a lower-dimensional representation, e.g. $D_s = 16$ followed by reranking with a higher capacity representation, e.g. $D_r = 2048$. In real-world scenarios where top ranking performance is the key objective, measured with $\text{mAP}@k$ where k covers a limited yet crucial real-estate, AR provides significant compute and memory gains over single-shot retrieval with representations of fixed dimensionality. Finally, the most expensive part of AR, as with any retrieval pipeline, is the nearest neighbour search for shortlisting. For example, even naive re-ranking of 200 images with 2048 dimensions only costs 400 KFLOPs. While we report exact search cost per query for all AR experiments, the shortlisting component of the pipeline can be sped-up using ANNS (HNSW). Appendix I has a detailed discussion on compute cost for exact search, memory overhead of HNSW indices and wall-clock times for both implementations. We note that using HNSW with 32 neighbours for shortlisting does not decrease accuracy during retrieval.

Figure 8 showcases the compute-vs-accuracy trade-off for adaptive retrieval using Matryoshka Representations compared to single-shot using fixed features with ResNet50 on ImageNet-1K. We observed that all AR settings lied above the Pareto frontier of single-shot retrieval with varying representation sizes. In particular for ImageNet-1K, we show that the AR model with $D_s = 16$ & $D_r = 2048$ is as accurate as single-shot retrieval with $d = 2048$ while being $\sim 128\times$ more efficient in theory and $\sim 14\times$ faster in practice (compared using HNSW on the same hardware). We show similar trends with ImageNet-4K, but note that we require $D_s = 64$ given the increased difficulty of the dataset. This results in $\sim 32\times$ and $\sim 6\times$ theoretical and in-practice speedups respectively. Lastly, while $K = 200$ works well for our adaptive retrieval experiments, we ablated over the shortlist size k in Appendix K.2 and found that the accuracy gains stopped after a

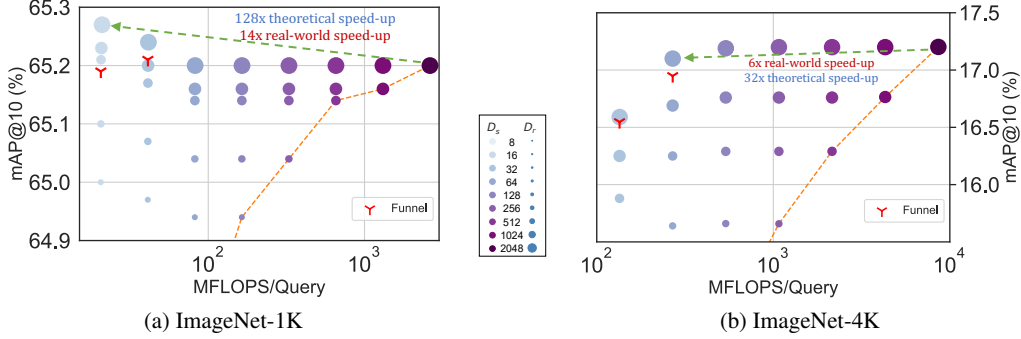


Figure 8: The trade-off between mAP@10 vs MFLOPs/Query for Adaptive Retrieval (AR) on ImageNet-1K (left) and ImageNet-4K (right). Every combination of D_s & D_r falls above the Pareto line (orange dots) of single-shot retrieval with a fixed representation size while having configurations that are as accurate while being up to 14 $\times$ faster in real-world deployment. Funnel retrieval is almost as accurate as the baseline while alleviating some of the parameter choices of Adaptive Retrieval.

point, further strengthening the use-case for Matryoshka Representation Learning and adaptive retrieval.

Even with adaptive retrieval, it is hard to determine the choice of D_s & D_r . In order to alleviate this issue to an extent, we propose **Funnel Retrieval**, a consistent cascade for adaptive retrieval. Funnel thins out the initial shortlist by a repeated re-ranking and shortlisting with a series of increasing capacity representations. Funnel halves the shortlist size and doubles the representation size at every step of re-ranking. For example on ImageNet-1K, a funnel with the shortlist progression of 200 $\rightarrow$ 100 $\rightarrow$ 50 $\rightarrow$ 25 $\rightarrow$ 10 with the cascade of 16 $\rightarrow$ 32 $\rightarrow$ 64 $\rightarrow$ 128 $\rightarrow$ 256 $\rightarrow$ 2048 representation sizes within Matryoshka Representation is as accurate as the single-shot 2048-dim retrieval while being $\sim 128\times$ more efficient theoretically (see Appendix F for more results). All these results showcase the potential of MRL and AR for large-scale multi-stage search systems [15].

5 Further Analysis and Ablations

Robustness. We evaluate the robustness of the MRL models trained on ImageNet-1K on out-of-domain datasets, ImageNetV2/R/A/Sketch [72, 34, 35, 94], and compare them to the FF baselines. Table 17 in Appendix H demonstrates that Matryoshka Representations for classification are at least as robust as the original representation while improving the performance on ImageNet-A by 0.6% – a 20% relative improvement. We also study the robustness in the context of retrieval by using ImageNetV2 as the query set for ImageNet-1K database. Table 9 in Appendix E shows that MRL models have more robust retrieval compared to the FF baselines by having up to 3% higher mAP@10 performance. This observation also suggests the need for further investigation into robustness using nearest neighbour based classification and retrieval instead of the standard linear probing setup. We also find that the zero-shot robustness of ALIGN-MRL (Table 18 in Appendix H) agrees with the observations made by Wortsman et al. [96]. Lastly, Table 6 in Appendix D.2 shows that MRL also improves the cosine similarity span between positive and random image-text pairs.

Few-shot and Long-tail Learning. We exhaustively evaluated few-shot learning on MRL models using nearest class mean [79]. Table 15 in Appendix G shows that that representations learned through MRL perform comparably to FF representations across varying shots and number of classes.

Matryoshka Representations realize a unique pattern while evaluating on FLUID [92], a long-tail sequential learning framework. We observed that MRL provides up to 2% accuracy higher on novel classes in the tail of the distribution, without sacrificing accuracy on other classes (Table 16 in Appendix G). Additionally we find the accuracy between low-dimensional and high-dimensional representations is marginal for pretrain classes. We hypothesize that the higher-dimensional representations are required to differentiate the classes when few training examples of each are known. This results provides further evidence that different tasks require varying capacity based on their difficulty.

Disagreement across Dimensions. The information packing in Matryoshka Representations often results in gradual increase of accuracy with increase in capacity. However, we observed that

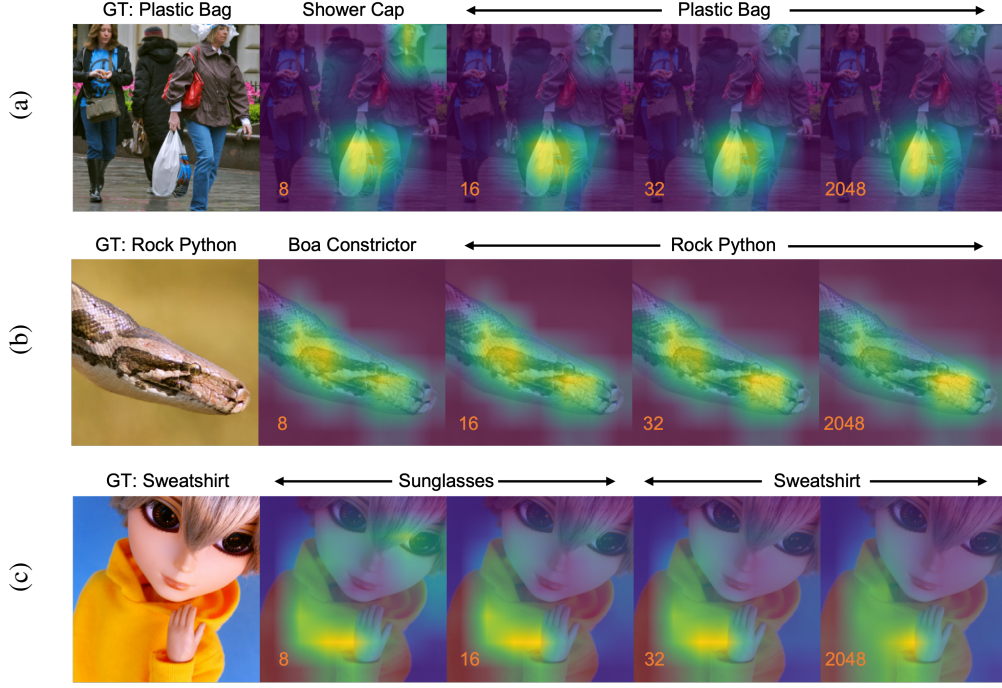


Figure 9: Grad-CAM [80] progression of predictions in MRL model across 8, 16, 32 and 2048 dimensions. (a) 8-dimensional representation confuses due to presence of other relevant objects (with a larger field of view) and predicts “shower cap” ; (b) 8-dim model confuses within the same super-class of “boa” ; (c) 8 and 16-dim models incorrectly focus on the eyes of the doll (“sunglasses”) and not the “sweatshirt” which is correctly in focus at higher dimensions; MRL fails gracefully in these scenarios and shows potential use cases of disagreement across dimensions.

this trend was not ubiquitous and certain instances and classes were more accurate when evaluated with lower-dimensions (Figure 12 in Appendix J). With perfect routing of instances to appropriate dimension, MRL can gain up to 4.6% classification accuracy. At the same time, the low-dimensional models are less accurate either due to confusion within the same superclass [24] of the ImageNet hierarchy or presence of multiple objects of interest. Figure 9 showcases 2 such examples for 8-dimensional representation. These results along with Appendix J put forward the potential for MRL to be a systematic framework for analyzing the utility and efficiency of information bottlenecks.

Superclass Accuracy. As the information bottleneck becomes smaller, the overall accuracy on fine-grained classes decreases rapidly (Figure 3). However, the drop-off is not as significant when evaluated at a superclass level (Table 24 in Appendix J). Figure 10 presents that this phenomenon

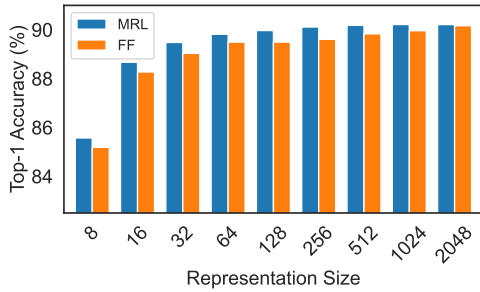


Figure 10: 31-way ImageNet-1K superclass classification across representation size for MRL & FF models showing the capture of underlying hierarchy through tight information bottlenecks.

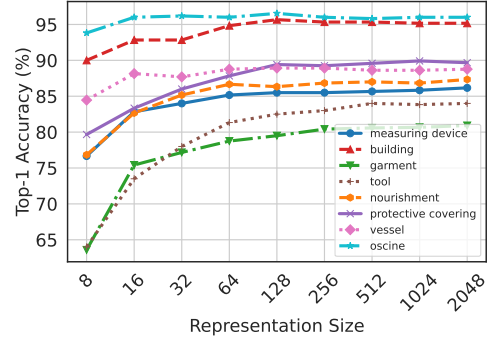


Figure 11: Diverse per-superclass accuracy trends across representation sizes for ResNet50-MRL on ImageNet-1K.